SV Sign: A Southern Dialect Multi view Sign Language Dataset and Spatio-Temporal Edge AI Evaluation

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Abstract. This study introduces SV Sign, a multi view isolated sign language dataset combined with a spatiotemporal Edge AI architecture evaluation system, focusing on the Southern dialect within public administration and frontline healthcare contexts. To bridge the regional data gap, the SV Sign stereoscopic dataset, comprising 16,200 raw video samples across 300 core vocabulary words, was constructed, within which this paper conducts a detailed evaluation on a subset of 51 gesture classes. To address the spatiotemporal representation problem and optimize edge computing, an Adaptive Time series Downsampling (ATD) algorithm is applied to reduce the sequence from 150 to 30 keyframes, optimizing the feature storage size to under 15 kilobytes per sample. Concurrently, the hybrid CNN LSTM network architecture is fine tuned via a cyclic graph flattening technique. Post training quantization reduces the binary Edge AI model size to 0.55 MB, equivalent to a compression ratio of 89.99%. Experimental results demonstrate that the system achieves an accuracy of 68.40%, maintaining a processing speed of 12.31 frames per second (FPS) on a single threaded CPU and 35 FPS with a latency of less than 30 milliseconds on a GPU platform. These metrics validate the feasibility of the proposed solution for deploying distributed AI systems on web platforms and low power mobile devices.

Keywords: SVSign Dataset · SpatioTemporal Modeling · Edge AI · Multiview Gesture Corpus · Model Quantization · Adaptive Temporal Downsampling

1 Introduction

Sign language remains the foundational communication medium for the deaf and hardofhearing, yet its proficiency among the hearing population is critically low [1, 2], driving the demand for automated isolated vocabulary translation [2, 3]. Although deep hybrid CNN RNN and attention networks successfully capture spatiotemporal features [2, 4], their practical deployment in Vietnamese public administration and healthcare sectors [1, 2] is heavily bottlenecked by dialect localization and edge infrastructure constraints [2, 10].

The localization barrier is exacerbated by single view datasets, such as MS ASL [5], WLASL [6], and DEVISIGN [7], which suffer from spatial projection flattening and trajectory confusion [2]. While the multi view VSL400 dataset resolves spatial occlusions, it strictly covers the Northern sign system [2]; deploying it in Southern regions triggers a severe data domain shift driven by regional variations in gesture velocity and palm morphology. Computationally, processing uncompressed 150 frame video sequences exhausts edge resources [2, 8–10], whereas centralized cloud architectures introduce severe identity privacy risks and network latency through raw video offloading [2, 10, 17]. Mitigating these limits demands edge based landmark extraction and optimized time series compression [10, 11].

To address these barriers, this paper proposes an edge cloud hierarchical isolated sign language recognition system featuring three core technical contributions:

- The study introduces a multi view Southern dialect dataset by compiling a localized 300 word corpus and performing a rigorous empirical evaluation across a 51 class public service and frontline healthcare subset to validate localization accuracy.
- The architecture incorporates an Adaptive Time series Downsampling (ATD) algorithm that utilizes a velocity driven keyframe selection mechanism to compress 150 raw frames into 30 optimized tensor keyframes based on wrist dynamics.
- The pipeline establishes a privacy by design edge deployment scheme by architecting an on device subsystem that extracts 133 morphological landmarks to eliminate raw video uploads and leverages computation graph quantization to preserve real time execution bandwidth across resource constrained web and mobile platforms.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 details the methodology and the proposed ATD algorithm; Section 4 analyzes the experimental results; and Section 5 concludes the work and outlines future research directives.

2 Related Work

Vision based sign language recognition categorizes into raw pixel processing and morphological landmark modeling [2, 3]. Early 2D/3D CNNs like I3D isolated spatiotemporal features directly from raw frames [12], but suffered from heavy computational overhead and environmental noise [8, 10]. Modern approaches shift toward geometric landmark tracking via MediaPipe Holistic [11] to encode hand, body, and facial coordinates into real valued vectors [4, 10]. This eliminates visual noise, establishing compressed feature spaces that facilitate lightweight models for edge device deployment [4, 10]. Modeling long term temporal dependencies is vital for time varying gesture sequence classification [2, 9]. Time series analysis evolved from Hidden Markov Models to LSTMs [9], mitigating vanishing gradients while capturing multi frame semantic dependencies [9, 15]. Merging CNN spatial extraction with recurrent blocks forms the hybrid CNN LSTM paradigm [15]. While Transformers improve accuracy [16], hybrid CNN LSTMs maintain execution speed and memory advantages during edge deployment, though keyframe filtering remains essential to prevent recurrent memory saturation [2, 10]. Deep learning efficiency relies heavily on standardized datasets [2, 5]. Benchmarks like RWTH PHOENIX Weather [13], WMT SLT [14], MS ASL [5], and DEVISIGN [7] rely on a single frontal perspective, introducing occlusion and trajectory confusion [2]. The multi view VSL400 dataset resolves these spatial ambiguities via three synchronous viewpoints [2] but remains strictly limited to Northern sign variants [2]. The absence of Southern morphological variations, signing speeds, and facial expressions induces a severe data domain shift, necessitating a localized, multi view Southern dialect gesture corpus [2].

3 Proposed Methodology

3.1 Data Pipeline and Subject Independent Synchronization

To address the Southern dialect data gap identified in WACV 2025 baselines, this architecture establishes a multi view gesture corpus for public administration and frontline healthcare contexts [2]. The ingestion pipeline synchronously captures visual streams via a threecamera network positioned at frontal (0°), left oblique (45°), and right oblique (-45°) angles to resolve non coplanar spatial trajectories and mitigate landmark occlusion [2]. The total dataset volume is mathematically defined in Equation (1):

$$D_{\text{total}} = C \times S \times \mathbb{R} \times V = 300 \times 6 \times 3 \times 3 = 16.200 \tag{1}$$

Within this formulation, D_{total}represents the aggregate video sample repository, mathematically determined by the initial vocabulary size C (C = 300), the number of physical signers S (S = 6), the repetition count per action R (R = 3), and the synchronized stereoscopic viewpoints V (V = 3) [2]. Out of this baseline matrix containing 16,200 independent captures, a targeted subset of 51 core vocabulary classes mapping directly to public administration and frontline healthcare contexts is isolated to drive network optimization and empirical evaluation protocols [2]. where D_{total} denotes the aggregate video sample repository, $C = 300$ is the vocabulary size, $S = 6$ represents the physical signers, $\mathbb{R} = 3$ is the repetition count per action, and $V = 3$ signifies the synchronized viewpoints [2]. From this baseline matrix of 16,200 independent captures, a targeted subset of 51 core vocabulary classes is isolated to drive network optimization and empirical evaluation protocols [2].

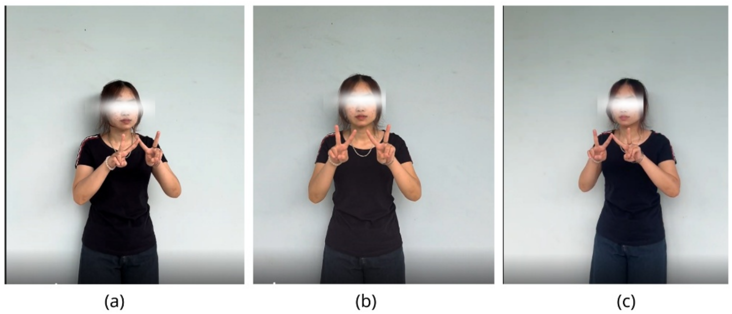

Fig. 1. Multi view data sample in the proposed dataset

Fig. 1 illustrates the multi view acquisition scheme, where the left oblique (a), frontal (b), and right oblique (c) perspectives cooperate to resolve geometric occlusion and projection flattening by capturing non coplanar spatial trajectories. Post acquisition, all streams are normalized to 30 FPS and categorized using a standardized string encoding convention [2]. A global stratified shuffling strategy mitigates morphological overfitting and spatial biases [2, 20], partitioning the unified dataset into an 80% training set, 10% validation set, and 10% testing set.

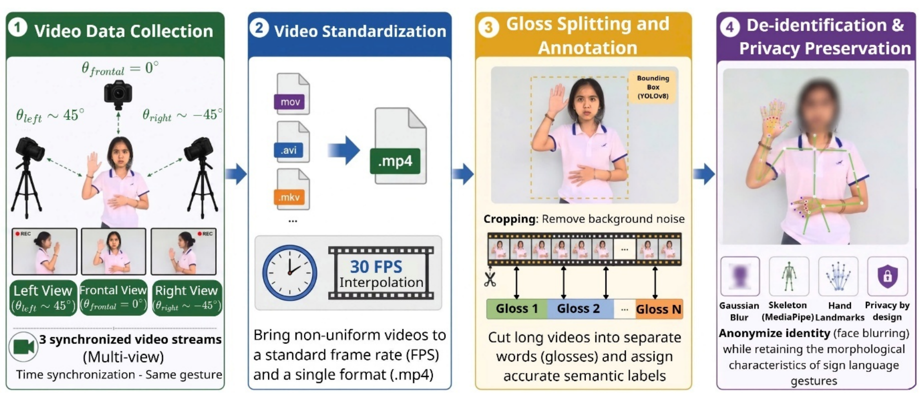

Fig. 2. Four stage closed loop workflow diagram comprising multi view data acquisition, frame rate normalization, lexical segmentation and filtering, and secure anonymization.

As shown in Fig. 2, the unidirectional standardization pipeline routes raw stereoscopic captures (Stage 1) into automated serialization using unified MP4 wrappers and a 30 FPS temporal filter [2]. The lexical segmentation phase (Stage 3) then leverages body bounding box detection to isolate gloss trajectories while attenuating environmental background noise. Finally, a face detection and structural de identification layer (Stage 4) applies localized Gaussian blurring to enforce privacy by design compliance without degrading high frequency skeletal landmark micro motion coordinates [2, 11].

3.2 Client Side Landmark Extraction and Privacy by Design

To enforce privacy by design, visual feature extraction is decentralized to client terminals [10, 17] using MediaPipe [11] for structural alignment with WACV 2025 benchmarks [2]. The pipeline digitizes 133 skeletal landmarks (33 pose, 42 hand, and 58 facial nodes) [10, 11], flattening 3D Cartesian coordinates into a single frame vector defined by Equation (2):

$$F_{\text{vector}} = L_{\text{total}} \times D_{\text{coord}} = 133 \times 3 = 399 \tag{2}$$

where L_{total} represents total body landmarks and D_{coord}signifies the three spatial axes. Immediately post extraction, raw video streams are permanently purged from edge memory to eliminate data leakage risks [17, 18].

3.3 Adaptive Temporal Downsampling Algorithm

To mitigate memory overhead from long, non uniform sequences and redundant static frames, an adaptive temporal downsampling (ATD) algorithm is integrated into the preprocessing pipeline [2, 10]. The solution operates by tracking the instantaneous kinematic velocity of both wrists. Given the spatial coordinates of a wrist landmark at frame t as$W_t = (x_t, y_t, z_t) \in \mathbb{R}^3$, the differential displacement velocity v_t between consecutive frames is determined via the Euclidean distance in Equation (3):

$$v_t = \sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2 + (z_t - z_{t-1})^2} \tag{3}$$

Here, $v_t$defines the frame wise velocity derived from current x_t, y_t, z_t and preceding $x_{t-1}, y_{t-1}, z_{t-1}$ 3D wrist coordinates. To isolate peak kinetic transitions, the keyframe selection weight P_t is normalized against the aggregate velocity of the N-frame sequence via Equation (4):

$$P_t = \frac{v_t}{\sum_{j=1}^{N} v_j} \tag{4}$$

In this distribution formulation, W_t is the retention probability value of frame t.The variable L represents the total number of initial raw frames of the input gesture video stream. The variable i is the running index of the summation operator representing each sequential frame and v_i is the displacement velocity at frame i. Based on this probability distribution, the filtering algorithm downsamples the initial sequence of 150 raw frames to exactly 30 keyframes containing the highest information density [2]. Following downsampling, the geometric coordinate arrays are packed into a fixed flat matrix as a feature space represented by the Tensor structure $TensorM$described in Equation (5):

$$M \in \mathbb{R}^{30\times 399} \tag{5}$$

In this spatial expression, M represents the morphological feature matrix representing a compressed gesture sample. The real number set symbol $\mathbb{R}$ defines the multidimensional vector space. The constant 30 is the number of keyframes retained by the algorithm, while 399 is the total number of spatial coordinate parameters computed from the 133 body morphological landmarks in Equation (2).

3.4 Data Packaging and Tensor Serialization

The serialization phase exports normalized samples directly into memory instead of large video files [19], limiting individual footprints under 15 KB [2, 17]. These blocks are loaded as a consolidated input tensor:

$$T \in \mathbb{R}^{B\times 30\times 399} \tag{6}$$

where B, 30, and 399 denote the batch size, synchronized frame length, and single view landmarks, respectively [2]. Under multi view setups, three camera matrices are concatenated in parallel prior to forward propagation [2]. This tensor feeds a hybrid CNN LSTM network [15, 20], where recurrent layers undergo loop unrolling along the temporal axis to flatten the computation graph into a static feedforward architecture [9, 15]. This transformation eliminates specialized hardware driver dependencies, accelerates loss processing, and establishes an optimized mathematical foundation for post training quantization pipelines targeting mobile deployment [19, 20].

4 Experiments

4.1 Experimental Setup and Hyperparameters

To evaluate cloud training against real time edge execution , the initial distributed phase leverages an Ada Lovelace architecture cloud cluster with an NVIDIA L4 GPU (24 GB VRAM) for accelerated spatiotemporal matrix operations. The software pipeline integrates TensorFlow and Keras alongside MediaPipe for landmark extraction and NumPy for multidimensional array linear algebra [11, 19]. To secure a stable convergence trajectory and suppress exploding gradients, the Adam optimizer is activated at a fixed learning rate of 10^{-4} [20].

Table 1. Configuration Setup of the CNN LSTM Hybrid Network and Experimental Environment

System Component	Parameter / Detailed Configuration	Role and Experimental Significance
Hardware Accelerator	NVIDIA L4 GPU 24 GB	Accelerates parallel spatio temporal matrix multiplication operations.
Core Deep Learning Framework	TensorFlow v2.15.0/Keras	Builds data pipelines and manages static computation graphs.
Optimization Algorithm (Optimizer)	Adam	Prevents gradient explosion, ensuring a smooth convergence trajectory..
Initial Learning Rate (LR)	10^{-4}	Prevents gradient explosion, ensuring a smooth convergence trajectory
Loss Function	Categorical Cross Entropy	Measures probability distribution distance across 51 gloss classes.
Batch Size	32	Fits hardware cache, optimizing data loading streams.
Total Epochs	40	Comprehensively monitors convergence trajectory (Optimal model at Epoch 35).
Dropout Rate	0.3 At recurrent hidden layers	Prevents overfitting and enhances generalization capability.
Activation Function	ReLU/Softmax	Introduces feature non linearity and normalizes output classification probabilities.

The configuration settings in Table 1 steer the mathematical convergence trajectory and eliminate systematic errors. Activating a fixed learning rate of 10^{-4} with the Adam optimizer thoroughly suppresses exploding gradients during backpropagation through recurrent hidden layers. Concurrently, a batch size of 32, categorical cross entropy loss, and a 0.3 dropout rate within the stacked LSTM layers efficiently exploit the physical cache bandwidth of the NVIDIA L4 hardware, preventing overfitting while ensuring real time inference throughput on edge devices.

4.2 Dataset Specification and Evaluation Protocol

The quantitative performance evaluation is executed on the independently constructed multiview Southern dialect gesture dataset named SV Sign Corpus. Within the experimental scope of this paper, the system focuses on the cross classification of 51 core vocabulary categories belonging to the public administration and frontline healthcare sectors.

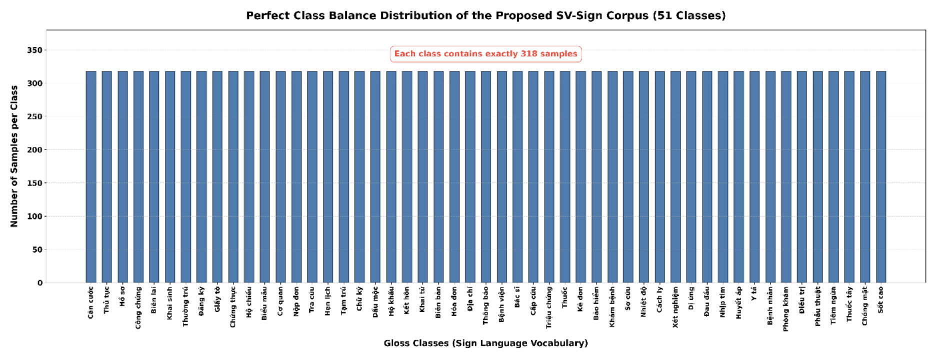

Fig. 3. Data balance chart of the 51 vocabulary classes

The detailed dataset specification regarding the scale distribution is visually illustrated through the frequency distribution chart in Fig. 3. The graphical indicators reveal a state of perfect class balance across the entire space of 51 target vocabulary classes, with no occurrence of the class imbalance phenomenon which is inherently the core factor causing classification bias for deep learning classifiers. The uniformity of sample density across each vocabulary item from the public administration sector to the frontline healthcare sector ensures that the categorical cross entropy loss function is optimized fairly across all target vector directions, establishing a standardized and reliable data foundation for the system to achieve the most optimal Southern dialect localization performance.

Table 2. Distribution of Target Vocabulary Categories within the SV Sign Corpus Gesture Dataset

Service Context Subsystem	**Class Count**	**Representative Target Vocabulary Examples**
Public Administration Sector	26 Vocabulary Classes	Identity Card, Procedure, Profile, Notarization, Receipt, Birth Certificate, Registration
Frontline Healthcare Sector	25 Vocabulary Classes	Hospital, Doctor, Emergency, Symptom, Medicine, Medical Examination, First Aid

As detailed in Table 2, the corpus provides a balanced coverage of 51 core isolated vocabulary classes split between public administration and frontline healthcare sectors. This localized dictionary presents a high classification difficulty due to narrow overlapping temporal movement trajectories and high similarity in static palm morphology. Consequently, the specialized dataset establishes a robust foundation to validate Southern dialect localization while proving the practical utility of the architecture for edge deployment within the community To ensure mathematical objectivity for the benchmarking scenarios, a global stratified shuffling strategy is applied to partition the comprehensive 16,200 sample matrix dataset into isolated partitions. This partitioning process is specifically defined according to the additive formulation in Equation (7):

$$D_{\text{total}} = D_{\text{train}} + D_{\text{val}} + D_{\text{test}} = 12{,}960 + 1{,}620 + 1{,}620 \quad (7)$$

In this partitioning formulation, the variable D_{total} represents the total size of the input flat matrix dataset, reaching 16,200 samples. The variable D_{train} is the data partition size dedicated to the training set, accounting for a proportion of 80 percent, equivalent to 12,960 samples, serving the adaptive machine learning phase. The variable D_{val} is the validation set, accounting for a proportion of 10 percent, equivalent to 1,620 samples, used to monitor the convergence trajectory. The variable D_{test} is the testing set, occupying the remaining 10 percent, which is completely isolated to evaluate the generalization capability of the mode.

Table 3. Quantitative statistics regarding the partitioning structure of the multiview gesture dataset after compression

Data Partition	Sample Quantity	Spatiotemporal Tensor Structure	Estimated Memory Capacity
Training Set	12.960 samples	Matrix of 12.960 by 30 by 399	Approximately 189.84 MB
Validation Set	1.620 samples	Matrix of 1.620 by 30 by 399	Approximately 23.73 MB
Testing Set	1.620 samples	Matrix of 1.620 by 30 by 399	Approximately 23.73 MB
Total Dataset	16.200 samples	Matrix of 16.200 by 30 by 399	Approximately 237.30 MB

Table 3 details the 80:10:10 data partition generated via global stratified shuffling, which enforces mathematical objectivity and eliminates subset data leakage. Driven by the ATD algorithm, the spatiotemporal matrix comprising 16,200 independent .npy samples is compressed to a footprint of 237.30 MB. This optimized storage scale resolves physical hardware I/O bottlenecks, enabling direct cloud cache loading while completely preventing VRAM graphics memory overflow during network optimization.

4.3 Cloud Based Large Scale Training

To validate multi view efficacy, a large scale evaluation compares single camera against three camera configurations on the Multi VSL dataset [2] across baseline architectures, including I3D [12], Video Swin Transformer [21], MViTv2 [22], and VTNPF [10]. Fig. 4 illustrates the empirical performance variations across the 200, 400, and 1000 vocabulary subsets [2].

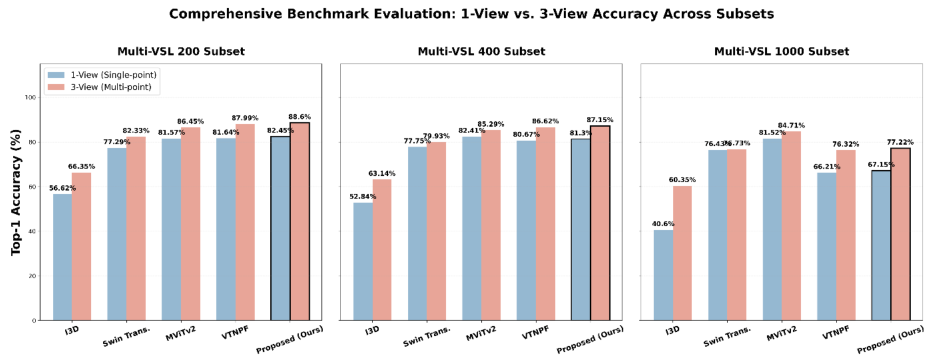

Fig. 4. Bar chart analyzing the performance cross comparison of classification accuracy between single camera and three camera configurations across baseline deep learning network architectures.

The multi view configuration consistently enhances classification accuracy across all VSL400 benchmark models [2]. Specifically, on the complex 1000 vocabulary subset, the three camera network elevates the I3D baseline performance from 40.60% to 60.35% [2]. Under the same challenging 1000 class scenario, the proposed hybrid CNN LSTM architecture integrated with adaptive temporal downsampling significantly outperforms the baselines, scaling accuracy from 67.15% in the single camera setup to a peak of 77.22% within the multi view deployment.

4.4 Convergence Log and Cross Dataset Evaluation

The distributed training process on the cloud graphics infrastructure records the loss function variation trajectory accompanied by the growth of system accuracy, as illustrated in Fig. 5.

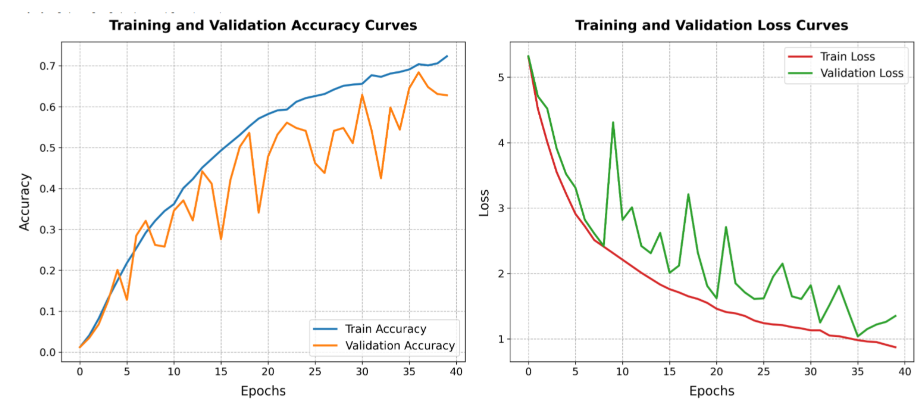

Fig. 5. Variation trajectory of accuracy and loss function during the system training and validation phases on the SV Sign Corpus dataset.

The hybrid CNN LSTM network converges at epoch 35, stabilizing validation accuracy at **68.40%** and reducing validation loss to **1.0385**. To evaluate domain shift phenomena, the model's classification performance on the proposed Southern dialect dataset is cross compared against standard baselines established on VSL400 [2], MS ASL [5], and WLASL [6].

Table 4. Comparative results of cross classification performance among methods across datasets

Classification System	Evaluation Dataset	Dialectal Context / Characteristics	Achieved Accuracy
Dinh et al. [2]	VSL400 Dataset [2]	Northern Dialect	71.24% [2]
Dinh et al. (re-impl.)	SV Sign Corpus (Proposed)	Cross dialect evaluation (Southern dialect)	41.15%
Hybrid CNN LSTM [15]	MS ASL Dataset [5]	Isolated American Sign Language	63.18% [5]
Hybrid CNN LSTM [15]	WLASL Dataset [6]	Large scale American Sign Language	60.52% [6]
Proposed Pipeline	SV Sign Corpus (Proposed)	Localized Southern dialect	68.40%

According to Table 4, evaluating the Northern dialect VSL400 [2] model directly on the Southern dialect dataset drops accuracy to 41.15%, confirming a severe regional domain shift driven by variations in finger morphology and movement velocity. Conversely, combining data localization with the proposed adaptive frame extraction algorithm yields a peak accuracy of 68.40% across the 51 localized healthcare and public administration classes.

4.5 Qualitative Analysis and Local Confusion

Behavioral analysis indicates that gestures with large spatial trajectories and wide amplitudes, such as Emergency, Hospital, and Isolation, consistently achieve peak classification accuracies between 95% and 98% because their continuous chest to head macro movements yield highly distinct dynamic features for the CNN layers [15]. Conversely, single camera configurations exhibit elevated confusion rates for sign pairs sharing identical static hand morphology but differing along the depth axis [2, 10]. Transitioning to the proposed multi view framework successfully mitigates this spatial bottleneck, leveraging peripheral perspectives to isolate these critical differential depth displacement vectors [2].

Table 5. Local confusion rates for ambiguous gloss pairs and error reduction margins under single view versus multi view configurations.

Ambiguous Gloss Pairs	Single View Error Rate	Multi View Error Rate	Error Reduction Margin
Document / Procedure	77.78%	44.44%	33.34%
Notarization / Authentication	66.67%	22.22%	44.45%
Form / Receipt	75.00%	37.50%	37.50%
Papers / Registration	55.56%	0.00%	55.56%

Table 5 presents a granular error analysis within the SV Sign Corpus, validating the 3 view paradigm over the 1 view baseline. The single view setup suffers from severe local confusion, peaking at a 77.78% error rate for the Document / Procedure pair due to 2D spatial projection flattening [2]. Conversely, the 3 view configuration achieves substantial error reductions ranging from 33.34% to 55.56% [2, 10], completely eliminating the Papers / Registration confusion by dropping its error rate from 55.56% to 0.00%. This quantitative leap demonstrates that synchronized lateral angles (45° left/right) effectively capture vital depth cues and resolve hand to body occlusions [2].

4.6 Hardware Serialization and Mobile Edge Optimization

To enable edge deployment and bypass proprietary cloud operator locks, the computation graph undergoes loop unrolling across all hidden layers [9, 15], transforming temporal recursive loops into a mobile compatible static feedforward structure. Subsequent post training quantization from 32 bit floating point to 8 bit integers reduces the model size by 89.99% (from 5.53 MB to 0.55 MB). Benchmarking in a single threaded vCPU environment records a latency of 81.25 ms per gesture (12.31 FPS), satisfying natural human signing speeds of 4, 5 words per second. Under parallel GPU acceleration, execution latency drops below 30 ms around 35 FPS, confirming the architecture's readiness for direct integration into standalone mobile applications.

5 Conclusion

This study presents an edge cloud hierarchical architecture for multi view isolated sign language recognition, mitigating regional dialect domain shifts and edge hardware constraints. The framework establishes a specialized Southern dialect dataset spanning 51 healthcare and public administration classes derived from 300 core vocabulary items. Algorithmically, adaptive temporal downsampling compresses 150 raw frames into 30 keyframes, reducing the feature storage footprint below 15 KB, while graph flattening and post training quantization shrink the model binary to 0.55 MB. The architecture yields a 68.40% validation accuracy, achieving real time inference speeds of 12.31 FPS on CPU and 35 FPS on GPU with sub 30 ms latencies, confirming its viability for resource constrained web and mobile deployment. Future research will focus on three core objectives: (1) expanding the pipeline to accommodate continuous sign language recognition; (2) integrating large language models to facilitate contextual translation of complex sentence structures; and (3) optimizing post quantization cross platform deployment to scale practical applications in public services and frontline healthcare.

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